

DES Numerically :-

8 Bit PT (10010111)

$K = 1010000010$

IP [2 6 3 4 1 5 7 8]

01011101

$K_1 \rightarrow$ Round 1

(1010010) $\downarrow$ 11011010

$K_2 \rightarrow$ Round 2

(01000011) $\downarrow$ 10100010

swap

00101010

IP-1

[4 1 3 5 7 2 8 6]

00111000

CT (8-bit)

(Ans) (38)

0101 1234 1101

LE₀ RE₀

1234 41232841

11101011

$K_1 \rightarrow \oplus$

Confusion

S-Box

0100 1111

1111

Diffusion

P-Box [2 4 8 1]

1111

RE₁

(1010)

LE₁ (1101)

1101 1234 1010

LE₁ RE₁

1234 01010101

EP

$K_2 \rightarrow \oplus$

S-Box

1111

P-Box

1111

RE₂

0010

LE₂ 1010

→ DES is a symmetric encryption algorithm with 16 rounds of operation.

→ For generating the key for each round, we have to perform left circular shift operation. Rounds 1, 2, 9 & 16 uses single shift & rest of the other rounds uses 2 shifts.

Ex: $\begin{array}{c} \overbrace{1101} \\ \downarrow 1\text{-bit Lcs} \\ 1011 \end{array}$

$\begin{array}{c} \overbrace{1101} \\ \downarrow 2\text{-bit Lcs} \end{array}$

$1011 \Rightarrow 0111$

↓

Rounds other than 1, 2, 9, 16

Advantages of DES:-

- It helps a well structured design which is easy to implement and very faster to operate in hardware.
- DES is a standard cryptographic algorithm that is the foundation for other advance techniques.
- DES is based on 2 important cryptographic concepts, i.e. Confusion & Diffusion.

Confusion:- It is the mechanism that makes the relationship between PT, CT & encryption key as complex as possible. It is performed by using any of the substitution algorithmic techniques.

Diffusion:- It is the techniques for making every bit of PT important for encryption & for every bit of changes in PT there will be a corresponding change in CT. It is performed by using any of the transposition techniques.

⇒ Hence confusion means replacement & diffusion means rearrangement.

Disadvantages of DES:-

- DES uses a small secret key size of 56 bit that can easily be broken by modern system.
- Therefore, this algorithm is not secure against modern crypto analysis methods.
- This method is a outdated ^{version} with a small ^{block} size with 64 bit. It is not secure enough thereby replace by an advance version of AES (Advanced Encryption Standard).

AES:-

- It is again a symmetric encryption algorithm with the much higher block size of 128 bits. The size of the secret key is decided by the no. of rounds of operation in AES.

→

	Block size	No. of Rounds	(Secret) key size	Round key
AES-128	128b	10	128b	128b
AES-192	128b	12	192b	128b
AES-256	128b	14	256b	128b

b-bits
B-Bytes

- The advantage of AES is that due to larger & variable key size, it has stronger security & is more complex than DES.
- The computational cost of AES is higher, which is both an advantage as well as disadvantage.
- Unlike DES, AES operates on a 4x4 matrix structure called as state matrix.

$$8 \text{ bits} = 1 \text{ Byte}$$

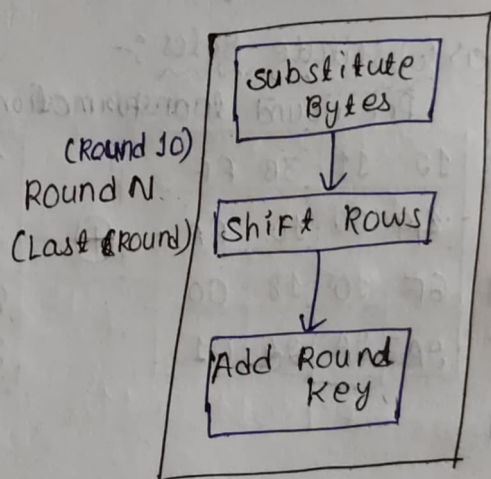
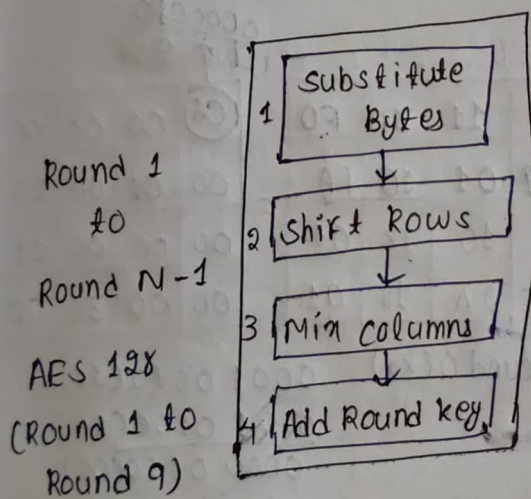
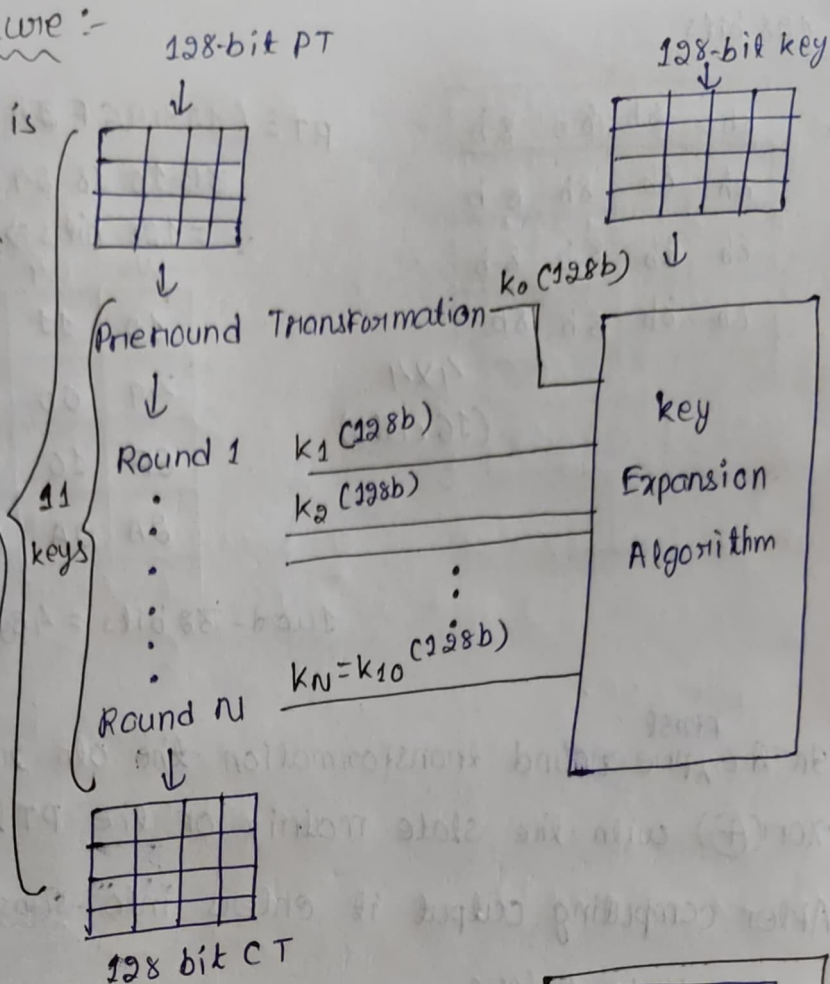
$$4 \text{ bits} = 1 \text{ nibble}$$

$$1 \text{ word} = 32 \text{ bits} = 4 \text{ Bytes}$$

AES Structure :-

Round key size is always 128b, innepative OF the AES.

AES-128
(128b PT,
128b K,
10 rounds
N=10)



- Depending upon the AES category there are 10, 12 or 14 rounds of operation.
- The data in each operation is arranged in a column major order in the AES state array.
- Every round has 4 different operation except the last round which does not have the mix column operation.

128 bits

8b	8b	8b	8b
8b	8b	8b	8b
8b	8b	8b	8b
8b	8b	8b	8b

4x4
(16 cells)

PT = (19 49 6F 9A 11 00 10 9A
3B 10 18 94 FF FA 00 01)_H
= 128 bits $\Rightarrow 32 \times 4$

w_0	w_1	w_2	w_3
19	11	3B	FF
49	00	10	FA
6F	10	18	00
9A	9A	94	01

1 word = 32 bits = 4 Byte

→ In the ^{first} pre round transformation the 0th round key is $\text{XOR}(\oplus)$ with the state matrix of the PT.

→ After computing output it enters into Substitute Bytes.

<a> Substitute Bytes :-
Pre round transformation

19	11	3B	FF
49	00	10	FA
6F	10	18	00
9A	9A	94	01

$\oplus$

19	11	3B	FF
49	01	10	FA
6F	10	18	01
9A	9A	12	01

00000010
row ↑ col

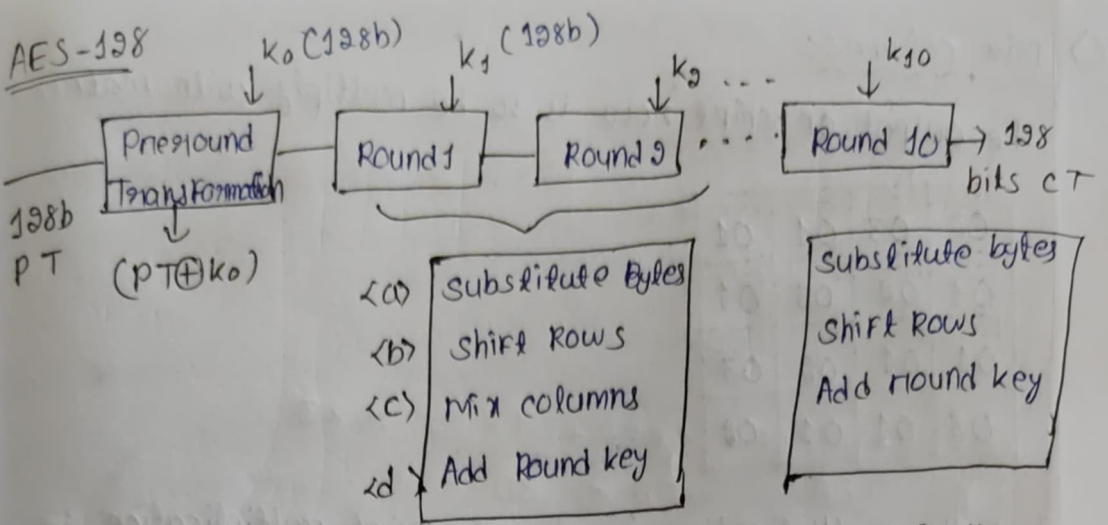
06	00	00	0F
00	01	00	00
00	00	00	01
00	00	36	00

Round 0(k_0)

0001 0010 (19)
0001 0100 (14)
0000 0110 (06)

→ In substitute byte the first 4 bit indicate the row no.
and next 4 bit indicates the column no.

→ The value at the row & column no. are searched in the 16 bit substitute byte box to generate the output state matrix.



Find the output after substitute by transformation if the input is

00 02 1A 2B 3C 4D FF F0
02 F0 FF 4D 3C 2B 1A 00

0 row 0 column

RC 00	3C	02	3C
02	4D	F0	RC 2B
1A	RC FF	FF	1A
2B	F0	4D	00

RC
2B

AES S-Box
(16x16)

1 row 1 col

63	EB	77	EB
77	E3	8C	F1
A9	16	16	A9
F1	8C	E3	63

4x4

Shift Row: In this transformation every row is shifted in a left circular manner

R0 → No shift
R1 → 1 LCS
R2 → 2 LCS
R3 → 3 LCS

63	EB	77	EB
E3	8C	F1	77
16	A9	A9	16
63	F1	8C	E3

16 16 A9 A9 → 1
16 A9 A9 16 → 2
8C E3 63 F1 → 1
E3 8C F1 8C → 2
63 F1 8C E3 → 3

<C> Mix columns:-

The output of shift now is to be multiply with matrix,

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

but the multiplication is a polynomial multiplication in Galois Field (2^3) that is different from general multiplication.

→ Whenever the power after multiplication is more than 7, then the polynomial is reduced after division with $(x^8 + x^4 + x^3 + x + 1)$.

$GF(2^3)$:-

In $GF(2^3)$ the set of values are 0 to 7.

$\{0, 1, 2, 3, 4, 5, 6, 7\}$

Represented as $\frac{x^7 \ x^6 \ x^5 \ x^4}{H1} \ \frac{x^3 \ x^2 \ x^1 \ x^0}{H2}$.

Q Represent $(1011 \ 0011)_b$ in a polynomial pattern.

$(\underline{1}011 \ 00\underline{11})_b$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $x^7 \ x^5 \ x^4 \quad x \ 1$

Polynomial - $(x^7 + x^5 + x^4 + x + 1)$

Q $FF \rightarrow 0111 \ 1111$

Q $AB \rightarrow 1010 \ 1011$

Polynomial $(x^6 + x^5 + x^4 + x^3 + x^2 + x + 1)$

Polynomial $(x^7 + x^5 + x^3 + x + 1)$

Q Calculate the hexadecimal equivalent for the following polynomial.

$$\rightarrow (x^6 + x + 1) \rightarrow 0100\ 0011 \rightarrow 43$$

$$\rightarrow (x^7 + x^6 + x^5) \rightarrow 1110\ 0000 \rightarrow E0$$

→ In Galois Field the concept of addition is similar to XOR ($\oplus$) operation & the subtraction ($-$) and division ($\div$) operation doesn't exist.

Ex: $F3 + F3 = 0$

$$A3 - F3 = A3 + F3$$

$$9 - 1 = 9 + (-1)$$

$$= 9 + 1$$

$$(11 * 91)_{GF} = 11 * 91$$

$$= 0001\ 0001 * 0010\ 0001$$

$$0001\ 0001 \rightarrow (x^4 + 1)$$

$$0010\ 0001 \rightarrow (x^5 + 1)$$

$$= (x^4 + 1) * (x^5 + 1)$$

$$= (x^9 + x^4 + x^5 + 1)$$

$$= (x^9 + x^5 + x^4 + 1)$$

$$x^8 + x^4 + x^3 + x + 1 \left| \begin{array}{l} x^9 + x^5 + x^4 + 1 \\ x^9 + x^5 + x^4 + x^3 + x + 1 \end{array} \right. x$$

$$(x^2 + x + 1)$$

$$= 0000\ 0111 \rightarrow (07)$$

→ $(11 * 91) = (07)$

Q Calculate

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

*
GF(2³)

Output of shift rows

04	00	00	00
BF	01	01	00
5D	00	00	01
30	00	00	01

	03	03	00

$$\frac{09 \times D4}{B3} + \frac{03 \times BF}{DA} + \frac{01 \times 5D}{5D} + \frac{01 \times 30}{30} = \underline{B3}$$

$$\begin{aligned} 09 \times D4 &= 0000\ 0010 \times 1101\ 0100 \\ &= (x) \times (x^7 + x^6 + x^4 + x^3) \\ &= x^8 + x^7 + x^5 + x^3 \end{aligned}$$

$$\begin{array}{l} x^8 + x^4 + x^3 + x + 1 \left) \begin{array}{l} x^8 + x^7 + x^5 + x^3 \\ x^8 + x^4 + x^3 + x + 1 \end{array} \left(\begin{array}{l} 1 \\ 1 \end{array} \right) \\ \hline (x^7 + x^5 + x^4 + x + 1) \rightarrow 1011\ 0011 \rightarrow B3 \end{array}$$

$$\begin{aligned} 03 \times BF &= 0000\ 0011 \times 1011\ 1111 \\ &= (x+1) \times (x^7 + x^5 + x^4 + x^3 + x^2 + x + 1) \\ &= x^8 + x^6 + x^5 + x^4 + x^3 + x^2 + x + x^7 + x^6 + x^4 + x^3 + x^2 + x + 1 \\ &= x^8 + x^7 + x^6 + 1 \end{aligned}$$

$$\begin{array}{l} x^8 + x^4 + x^3 + x + 1 \left) \begin{array}{l} x^8 + x^7 + x^6 + 1 \\ x^8 + x^4 + x^3 + x + 1 \end{array} \left(\begin{array}{l} 1 \\ 1 \end{array} \right) \\ \hline x^7 + x^6 + x^4 + x^3 + x \rightarrow 1101\ 1010 \\ \rightarrow DA \end{array}$$

$$= B3 \oplus DA \oplus 5D \oplus 30$$

Q calculate the output after shift row & mix column transformation if I/P =

00	00	00	00
00	00	00	00
00	12	00	00
00	00	00	01

00	00	00	00
00	00	00	00
00	00	00	12
01	00	00	00

12 00 00 00
00 00 00 12

→ shift row

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

*

00	00	00	00
00	00	00	00
00	00	00	12
01	00	00	00

=

01	00	00	12
01	00	00	$\frac{03 \times 12}{36} = 36$
03	00	00	$\frac{02 \times 12}{24} = 24$
02	00	00	12

$$02 \times 00 + 03 \times 00 + \frac{01 \times 00}{00} + \frac{01 \times 01}{01} (x)$$

$$\begin{aligned} 03 \times 12 &= 0011 \times 1100 \\ &= (x^5 + x^4) * (x^3 + x^2) \\ &= x^8 + x^7 + x^7 + x^6 \\ &= x^8 + x^6 \end{aligned}$$

$$\begin{array}{r} x^8 + x^4 + x^3 + x + 1 \\ x^8 + x^6 \\ \hline \end{array}$$

$$x^6 + x^4 + x^3 + x + 1 \rightarrow 01011011 \rightarrow 5B$$

$$\begin{aligned} 00000011 * 00010010 \\ &= (x+1) * (x^4+x) \\ &= x^5 + x^2 + x^4 + x \\ &= x^5 + x^4 + x^2 + x \rightarrow 00110110 \\ &\quad \xrightarrow{36} 36 \\ &\quad \xrightarrow{24} 24 \end{aligned}$$

$$02 \times 12 = 00000010 * 00010010$$

$$= (x+1) * (x^4+x)$$

$$= (x^5 + x^2) = 00100100 \rightarrow 24$$

$$\begin{array}{r} x^8 + x^4 + x^3 + x + 1 \\ x^5 + x^2 \\ \hline \end{array}$$

(d) Add Round key:-

In add round key, the output of mix column transformation is XOR ($\oplus$) with the round key of the corresponding column.

00 00 00 00
00 00 00 00

00	00	00	00
00	00	00	00
00	00	00	00
00	00	00	00

00 00 00 00

00	00	00	00
00	00	00	00
00	00	00	00
00	00	00	00

00	00	00	00
00	00	00	00
00	00	00	00
00	00	00	00

$\times$

10	10	00	00
10	10	00	00
00	00	10	10
00	00	10	10

$$(v) \frac{00 \times 10 + 00 \times 10 + 00 \times 00 + 00 \times 00}{10 \quad 00}$$

$$01001000 \times 11000000 = 01 \times 00$$

$$(x^2 + x) \times (1 + x) =$$

$$x^2 + x^2 + x + x =$$

$$01111000 \times 11001000 = 00 + x^2 + x^2 + 00 =$$

$$x^2 + x^2 + x^2 + x^2 =$$

$$00111100 \times 00111100 = 01 \times 00$$

$$(x^2 + x^2) \times (x^2 + x^2) =$$

$$x^2 + x^2 + x^2 + x^2 =$$

$$x^2 + x^2 =$$

$$1 \times x^2 + x^2 \times 1 = x^2 + x^2$$

$$11011010 \times 11011010 = 1 + x^2 + x^2 + x^2$$

$$1 + x^2 + x^2 + x^2 =$$

$$01001000 \times 01001000 = 01 \times 00$$

$$(x^2 + x^2) \times (x^2 + x^2) =$$

$$x^2 + x^2 + x^2 + x^2 =$$

$$x^2 + x^2 + x^2 + x^2 =$$